

Chapter 1 - Just enough category theory to be dangerous

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1 Categories and functors

Exercise 1.A

UNIMPORTANT EXERCISE.

A category in which each morphism is an isomorphism is called a groupoid. (This notion is not important in what we will discuss. The point of this exercise is to give you some practice with categories, by relating them to an object you know well.)

- (a) A perverse definition of a group is: a groupoid with one object. Make sense of this. (Similarly, in case you care: a perverse definition of a monoid is: a category with one object.)
- (b) Describe a groupoid that is not a group.

Solution. 1. Since there's only one object, namely e , in the category, all the morphisms in such category will send e to e .

- The combinations of morphisms are always associative.
- There must exist an identity morphism, which'll be the identity in the group.
- Since it's a groupoid, every morphism has a inverse.

(In fact every group can be represented using the language of category. Consider Cayley's Theorem and we can see every group element can be regarded as an automorphism, satisfying the case of morphism to a object itself.)

2. We consider a category $\mathcal{C}$ whose objects are A, B . Consider

$$\text{Mor}(\mathcal{C}) = \{\text{Id}_A, \text{Id}_B\}.$$

and we can see that it's a groupoid where the elements corresponding to Id_A, Id_B cannot have multiplication, therefore not a group.

Exercise 1.B

If A is an object in a category $\mathcal{C}$, show that the invertible elements of $\text{Mor}(A, A)$ form a group (called the automorphism group of A , denoted $\text{Aut}(A)$). What are the automorphism groups of the objects in Examples 1.1.2 and 1.1.3? Show that two isomorphic objects have isomorphic automorphism groups. (For readers with a topological background: if X is a topological space, then the fundamental groupoid is the category where the objects are points of X , and the morphisms $x \rightarrow y$ are paths from x to y , up to homotopy. Then the automorphism group of x_0 is the (pointed) fundamental group $\pi_1(X, x_0)$. In the case where X is connected, and $\pi_1(X)$ is not abelian, this illustrates the fact that for a connected groupoid — whose definition you can guess — the automorphism groups of the objects are all isomorphic, but not canonically isomorphic.)

Proof. 1. Similar to the case in **1.1.A**, we can use exactly the same proof to show that invertible elements of $\text{Mor}(A, A)$ forms a group.

2. Automorphism groups of objects in **Example 1.1.2** are those permutation groups permutating the elements in a group.

3. Automorphism groups of objects in **Example 1.1.3** are those invertible linear transformations mapping a linear space to itself. □

Exercise 1.C [Skipped, require 1.1.21 to solve.]

Let $(\cdot)^{\vee\vee} : f.d.Vec_k \rightarrow f.d.Vec_k$ be the double dual functor from the category of finite-dimensional vector spaces over k to itself. Show that $(\cdot)^{\vee\vee}$ is naturally isomorphic to the identity functor on $f.d.Vec_k$. (Without the finite-dimensionality hypothesis, we only get a natural transformation of functors from id to $(\cdot)^{\vee\vee}$.)

Proof. (i) In the case of finite cases, the set of bases of $Vec_k, Vec_k^\vee$ is finite. Therefore the functor $(\cdot)^\vee$ must be isomorphism. □

Exercise 1.D

Show that $\mathcal{V} \rightarrow f.d.Vec_k$ gives an equivalence of categories, by describing an “inverse” functor. (Recall that we are being cavalier about set-theoretic assumptions, see Caution 0.3.1, so feel free to simultaneously choose bases for each vector space in $f.d.Vec_k$. To make this precise, you will need to use Gödel-Bernays set theory or else replace $f.d.Vec_k$ with a very similar small category, but we won’t worry about this.)

Proof. Since it’s finite, just find out the bases and we can define a trivial isomorphism between $\mathcal{V}$ and $f.d.Vec_k$ by mapping those bases respectively. □

2 Universal properties determine an object up to unique isomorphism

Exercise 2.A

Show that any two initial objects are uniquely isomorphic. Show that any two final objects are uniquely isomorphic.

Proof. 1. For any two initial objects A, B , we define

$$\varphi : A \mapsto B, \psi : B \mapsto A.$$

And we can see

$$\varphi \circ \psi : B \mapsto B, \psi \circ \varphi : A \mapsto A.$$

Also, notice since A, B are initial objects, they have only one unique map to every objects in the category, including themselves. And Id_A, Id_B maps A, B to themselves. Hence we obtain

$$\begin{aligned} Id_A &= \psi \circ \varphi \\ Id_B &= \psi \circ \varphi. \end{aligned}$$

So we know φ, ψ are isomorphisms.

2. The proof is similar to (a), skipped. □

Exercise 2.B

What are the initial and final objects in *Sets*, *Rings*, and *Top* (if they exist)? How about in the two examples of §1.1.9?

Solution. 1. In the category of sets, the initial and final object is $\emptyset$ and the class of all sets, respectively.

2. In the category of rings, the initial and final object is the zero ring and the class of all rings, respectively.

3. Skipped (haven’t learn topology yet).

Exercise 2.C

Show that $A \rightarrow S^{-1}A$ is injective if and only if S contains no zerodivisors. (A zerodivisor of a ring A is an element a such that there is a nonzero element b with $ab = 0$. The other elements of A are called non-zerodivisors. For example, an invertible element is never a zerodivisor. Counter-intuitively, 0 is a zerodivisor in every ring but the 0 -ring. More generally, if M is an A -module, then $a \in A$ is a zerodivisor for M if there is a nonzero $m \in M$ with $am = 0$. The other elements of A are called non-zerodivisors for M . Equivalently, and very usefully, $a \in A$ is a non-zerodivisor for M if and only if $\times a : M \rightarrow M$ is an injection, or equivalently in the language of §1.5, if

$$0 \longrightarrow M \xrightarrow{\times a} M$$

is exact.)

Proof. • ($\Leftarrow$): When S contains no zerodivisors, we prove by contradiction: Assume that under the canonical ring map $\varphi : A \mapsto S^{-1}A$, there exist $a_1, a_2 \in A$, s.t. $\varphi(a_1) = \varphi(a_2)$. Now we have $a_1/1 = a_2/1$, which means

$$\exists s \in S, \text{ s.t. } s(a_1 - a_2) = 0.$$

Which contradicts with the fact that S has no zerodivisors!

• ($\Rightarrow$): When $\varphi : A \mapsto S^{-1}A$ is injective, we assume S contains a zerodivisors s at which

$$\exists d \in A, \text{ s.t. } d \neq 0, s \cdot d = 0.$$

Now let $a_1, a_2 \in A$, s.t. $a_1 - a_2 = d$. Since φ is injective, we must have $\varphi(a_1) \neq \varphi(a_2)$. Which contradicts with the fact that

$$s(a_1 - a_2) = s \cdot d = 0.$$

□

Exercise 2.D

Verify that $A \rightarrow S^{-1}A$ satisfies the following universal property: $S^{-1}A$ is initial among A -algebras B where every element of S is sent to an invertible element in B . (Recall: the data of “an A -algebra B ” and “a ring map $A \rightarrow B$ ” are the same.) Translation: any map $A \rightarrow B$ where every element of S is sent to an invertible element must factor uniquely through $A \rightarrow S^{-1}A$. Another translation: a ring map out of $S^{-1}A$ is the same thing as a ring map from A that sends every element of S to an invertible element. Furthermore, an $S^{-1}A$ -module is the same thing as an A -module for which $s \times \cdot : M \rightarrow M$ is an A -module isomorphism for all $s \in S$.

Proof. Suppose

$$\begin{aligned} f : A &\mapsto B \\ \varphi : A &\mapsto S^{-1}A. \end{aligned}$$

By the first translation in by author, we have to prove that there an unique ring homomorphism ψ mapping $S^{-1}A$ to B . Recall that $\forall a \in A, f(a) = \varphi(a/1)$. Also, for all $a \in A, s \in S$,

$$\frac{a}{s} \cdot \frac{s}{1} = \frac{a}{1}.$$

So

$$\begin{aligned} \psi\left(\frac{a}{s}\right) \cdot \psi\left(\frac{s}{1}\right) &= \psi\left(\frac{a}{1}\right) \\ \psi\left(\frac{a}{s}\right) &= \psi\left(\frac{a}{1}\right) \cdot \psi\left(\frac{s}{1}\right)^{-1} \end{aligned}$$

And we can see $\psi\left(\frac{a}{s}\right)$ is uniquely defined!

□

Exercise 2.E

Show that $\phi : M \rightarrow S^{-1}M$ exists, by constructing something satisfying the universal property. Hint: define elements of $S^{-1}M$ to be of the form m/s where $m \in M$ and $s \in S$, and $m_1/s_1 = m_2/s_2$ if and only if for some $s \in S$, $s(s_2m_1 - s_1m_2) = 0$. Define the additive structure by $(m_1/s_1) + (m_2/s_2) = (s_2m_1 + s_1m_2)/(s_1s_2)$, and the $S^{-1}A$ -module structure (and hence the A -module structure) is given by $(a_1/s_1) \cdot (m_2/s_2) = (a_1m_2)/(s_1s_2)$.

Proof. We follow the hint from author: Define

$$S^{-1}M = \left\{ \frac{m}{s} \mid m \in M, s \in S \right\}.$$

And $S^{-1}M$ meet the requirement to become a module by: $\forall m_1, m_2 \in M, s_1, s_2 \in S$:

$$\frac{m_1}{s_1} = \frac{m_2}{s_2} \iff \exists s \in S, s(s_2m_1 - s_1m_2) = 0.$$

$$\frac{m_1}{s_1} + \frac{m_2}{s_2} = \frac{m_1s_2 + m_2s_1}{s_1s_2}.$$

And the $S^{-1}A$ module structure

$$\forall a \in A, s_1, s_2 \in S, m \in M, \frac{a}{s_1} \cdot \frac{m}{s_2} = \frac{am}{s_1s_2}.$$

Also, we define $M \Rightarrow S^{-1}M$ by

$$m \longrightarrow \frac{m}{1}.$$

Now for any A -module maps $M \rightarrow N$, we'll prove ϕ satisfies the universal property, i.e. the map $S^{-1}M \rightarrow N$, we denote it by φ , is unique: $\forall \frac{m}{s} \in S^{-1}M$, notice:

$$\begin{cases} \alpha(m) \in N \\ \alpha(s)^{-1} \in N \end{cases}.$$

Therefore we have $\varphi(\frac{m}{s}) = \alpha(m)\alpha(s)^{-1}$. So the ϕ in the problem satisfy the uniqueness of universal property, so it exist. $\square$

Exercise 2.F (a) Show that localization commutes with finite products, or equivalently, with finite direct sums. In other words, if $M_1, \dots, M_n$ are A -modules, describe an isomorphism (of A -modules, and of $S^{-1}A$ -modules) $S^{-1}(M_1 \times \dots \times M_n) \rightarrow S^{-1}M_1 \times \dots \times S^{-1}M_n$.

(b) Show that localization commutes with arbitrary direct sums.

(c) Show that "localization does not necessarily commute with infinite products": the obvious map $S^{-1}(\prod_i M_i) \rightarrow \prod_i S^{-1}M_i$ induced by the universal property of localization is not always an isomorphism. (Hint: $(1, 1/2, 1/3, 1/4, \dots) \in \mathbb{Q} \times \mathbb{Q} \times \dots$)

Proof. 1. Intuitively, we can let

$$\frac{(m_1, m_2, \dots, m_n)}{s} \longrightarrow \left(\frac{m_1}{s}, \frac{m_2}{s}, \dots, \frac{m_n}{s} \right).$$

On the other hand, inductively we only have to consider the case $n = 2$, and we let

$$\left(\frac{m_1}{s_1}, \frac{m_2}{s_2} \right) \longrightarrow \left(\frac{m_1s_2}{s_1s_2}, \frac{m_2s_1}{s_1s_2} \right).$$

And we're done after induction.

2. Notice that in the case of direct sum, all but finitely many coordinates is zero. Therefore the proof is exactly the same as the case in (a).

3. Counterexample: consider group of modules $\{M_n\}_{n=1}^\infty$, and set of prime numbers $P = \{p_n\}_{n=1}^\infty$. We consider fraction of $\{M_n\}_{n=1}^\infty$ by positive integers, we cannot map $\frac{m_1}{p_1}, \frac{m_2}{p_2}, \dots$ to a proper element in $S^{-1}(\bigotimes_{k \in I} M_k)$ since $p_1, p_2, \dots$ cannot have a shared denominator. $\square$

Exercise 2.G

EXERCISE (IF YOU HAVEN'T SEEN TENSOR PRODUCTS BEFORE).

Show that $\mathbb{Z}/(10) \otimes_{\mathbb{Z}} \mathbb{Z}/(12) \cong \mathbb{Z}/(2)$. (This exercise is intended to give some hands-on practice with tensor products.)

Proof. More generally, we'll prove that

$$\forall m, n \in \mathbb{Z}^+, \mathbb{Z}/(m) \otimes_{\mathbb{Z}} \mathbb{Z}/(n) \cong \mathbb{Z}/(\text{Gcd}(m, n)).$$

First, we know $\mathbb{Z}/(m) \otimes_{\mathbb{Z}} \mathbb{Z}/(n)$ is generated by $1_{\mathbb{Z}/(m)} \otimes 1_{\mathbb{Z}/(n)}$. Which means that every elements of $\mathbb{Z}/(m) \otimes_{\mathbb{Z}} \mathbb{Z}/(n)$ can be written in the form of $k \cdot (1_{\mathbb{Z}/(m)} \otimes 1_{\mathbb{Z}/(n)})$, $k \in \mathbb{Z}^+$. Also, $\forall i, j \in \mathbb{Z}^+$,

$$k \cdot (1_{\mathbb{Z}/(m)} \otimes 1_{\mathbb{Z}/(n)}) = (mi + nj) \cdot k \cdot (1_{\mathbb{Z}/(m)} \otimes 1_{\mathbb{Z}/(n)}).$$

And the proof can be done by Euclidean algorithm. □

Exercise 2.H [Proved in Atiyah]

IMPORTANT EXERCISE: RIGHT-EXACTNESS OF $(\cdot) \otimes_A N$.

Show that $(\cdot) \otimes_A N$ gives a covariant functor $\text{Mod}_A \rightarrow \text{Mod}_A$. Show that $(\cdot) \otimes_A N$ is a right-exact functor, i.e., if

$$M' \rightarrow M \rightarrow M'' \rightarrow 0$$

is an exact sequence of A -modules (which means $f: M \rightarrow M''$ is surjective, and M' surjects onto the kernel of f ; see §1.5), then the induced sequence

$$M' \otimes_A N \rightarrow M \otimes_A N \rightarrow M'' \otimes_A N \rightarrow 0$$

is also exact. This exercise is repeated in Exercise 1.5.G, but you may get a lot out of doing it now. (You will be reminded of the definition of right-exactness in §1.5.6.)

Exercise 2.I [Trivial]

Show that $(T, t: M \times N \rightarrow T)$ is unique up to unique isomorphism. Hint: first figure out what “unique up to unique isomorphism” means for such pairs, using a category of pairs (T, t) . Then follow the analogous argument for the product.

Proof. Because from the textbook content right above this question, we see any two maps T, T' has a unique A -linear map to each other. □

Exercise 2.J

Show that the construction of §1.2.5 satisfies the universal property of tensor product.

Proof. Because the construction in §1.2.5 is a bilinear map, is unique up to isomorphism. □

Exercise 2.K

IMPORTANT EXERCISE.

(a) If M is an A -module and $A \rightarrow B$ is a morphism of rings, give $B \otimes_A M$ the structure of a B -module (this is part of the exercise). Show that this describes a functor $\text{Mod}_A \rightarrow \text{Mod}_B$.

(b) (tensor product of rings) If further $A \rightarrow C$ is another morphism of rings, show that $B \otimes_A C$ has a natural structure of a ring. Hint: multiplication will be given by $(b_1 \otimes c_1)(b_2 \otimes c_2) = (b_1 b_2) \otimes (c_1 c_2)$. (Exercise 1.2.U will interpret this construction as a fibered coproduct.)

Proof. 1. • $\forall b_1, b_2 \in B, b \otimes_A m \in B \otimes_A M$, we have

$$\begin{aligned} (b_1 + b_2)(b \otimes_A m) &= (b_1 + b_2)b \otimes_A m \\ &= (b_1 b + b_2 b) \otimes_A m \\ &= b_1 b \otimes_A m + b_2 b \otimes_A m \\ &= b(b_1 \otimes_A m) + b(b_2 \otimes_A m). \end{aligned}$$

$$\begin{aligned} (b_1 b_2)(b \otimes_A m) &= (b_1 b_2 b) \otimes_A m \\ &= b_1(b_2(b \otimes_A m)). \end{aligned}$$

$$b(b_1 \otimes_A m + b_2 \otimes_A m) = b(b_1 \otimes_A m) + b(b_2 \otimes_A m).$$

- Denote such map by $\mathcal{F}$. We see for ring A and $B \otimes_A M \in \text{Mod}_B$,

$$\mathcal{F} : A \mapsto B \otimes_A M.$$

And for $\varphi \in \text{Mor}(A)$, $\varphi : A \mapsto A'$,

$$\mathcal{F}(\varphi) : B \otimes_A M \mapsto B \otimes_{A'} M.$$

2. We define addition as formal sum and multiplication as

$$(b_1 \otimes c_1) \cdot (b_2 \otimes c_2) = (b_1 b_2) \otimes (c_1 c_2).$$

And we can easily exam the properties to let it becomes a ring. □

Exercise 2.L

IMPORTANT EXERCISE.

If S is a multiplicative subset of A and M is an A -module, describe a natural isomorphism $(S^{-1}A) \otimes_A M \xrightarrow{\sim} S^{-1}M$ (as $S^{-1}A$ -modules and as A -modules).

Proof. We can define

$$(S^{-1}A) \otimes_A M \longrightarrow S^{-1}M.$$

As

$$\left(\frac{a}{s}\right) \otimes m \mapsto \frac{am}{s}.$$

And we can easily check that it meets the requirement to become a isomorphism. □

Exercise 2.M

EXERCISE ($\otimes$ COMMUTES WITH $\oplus$).

Show that tensor products commute with arbitrary direct sums: if M and $\{N_i\}_{i \in I}$ are all A -modules, describe an isomorphism

$$M \otimes (\oplus_{i \in I} N_i) \xrightarrow{\sim} \oplus_{i \in I} (M \otimes N_i).$$

Proof. Because I only has finite nonzero elements, so the properties of tensor product can be verified easily. □

Exercise 2.N

EXERCISE (FIBERED PRODUCTS OF SETS).

Show that in Sets,

$$X \times_Z Y = \{(x, y) \in X \times Y : \alpha(x) = \beta(y)\}.$$

More precisely, show that the right side, equipped with its evident maps to X and Y , satisfies the universal property of the fibered product. (This will help you build intuition for fibered products.)

Proof. It's because

$$\alpha \circ \text{pr}_X = \beta \circ \text{pr}_Y.$$

□

Exercise 2.O

If X is a topological space, show that fibered products always exist in the category of open sets of X , by describing what a fibered product is. (Hint: it has a one-word description.)

Solution. Intersection.

Because morphisms are inclusion embeddings.

Exercise 2.P

If Z is the final object in a category $\mathcal{C}$, and $X, Y \in \mathcal{C}$, show that " $X \times_Z Y = X \times Y$ ": "the" fibered product over Z is uniquely isomorphic to "the" product. Assume all relevant (fibered) products exist. (This is an exercise about unwinding the definition.)

Proof. Denote the natural maps

$$\text{pr}_X : X \times Y \longrightarrow X.$$

$$\text{pr}_Y : X \times Y \longrightarrow Y.$$

And the unique map

$$\alpha : X \longrightarrow Z.$$

$$\beta : Y \longrightarrow Z.$$

Now we have to prove

$$\alpha \circ \text{pr}_X = \beta \circ \text{pr}_Y.$$

Which do work since Z is a final diagram, from which we know, up to isomorphisms, has only a single morphism sending $X \times Y$ to Z . $\square$

Exercise 2.Q

USEFUL EXERCISE: TOWERS OF CARTESIAN DIAGRAMS ARE CARTESIAN DIAGRAMS.

If the two squares in the following commutative diagram are Cartesian diagrams, show that the “outside rectangle” (involving U , V , Y , and Z) is also a Cartesian diagram.

$$\begin{array}{ccc} U & \longrightarrow & V \\ \downarrow & & \downarrow \\ W & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y & \longrightarrow & Z \end{array}$$

Proof. For notations, let

$$\begin{array}{ccc} U & \xrightarrow{f_1} & V \\ \downarrow g_1 & & \downarrow h_1 \\ W & \xrightarrow{f_2} & X \\ \downarrow g_2 & & \downarrow h_2 \\ Y & \xrightarrow{f_3} & Z \end{array}.$$

And we can see

$$\begin{aligned} h_2 \circ h_1 \circ f_1 &= h_2 \circ f_2 \circ g_1 \\ &= f_3 \circ g_2 \circ g_1. \end{aligned}$$

$\square$

Exercise 2.R

Given morphisms $X_1 \rightarrow Y$, $X_2 \rightarrow Y$, and $Y \rightarrow Z$, show that there is a natural morphism $X_1 \times_Y X_2 \rightarrow X_1 \times_Z X_2$, assuming that both fibered products exist. (This is trivial once you figure out what it is saying. The point of this exercise is to see why it is trivial.)

Proof.

$$\begin{array}{ccccc} X_1 \times_Y X_2 & & & & \\ & \searrow \exists! & & \searrow & \\ & X_1 \times_Z X_2 & \longrightarrow & X_2 & \\ & \downarrow & & \downarrow & \\ & X_1 & \longrightarrow & Y & \\ & & \searrow & \searrow & \\ & & & & Z \end{array}$$

Because from the problem we have the commutative diagram

$$\begin{array}{ccc} X_1 \times_Y X_2 & \longrightarrow & X_2 \\ \downarrow & & \downarrow \\ X_1 & \dashrightarrow & Z \end{array} .$$

And by the definition of fibered product, there must exist a unique factoring map $X_1 \times_Y X_2 \longrightarrow X_1 \times_Z X_2$. $\square$

Exercise 2.S

IMPORTANT EXERCISE: THE DIAGONAL-BASE-CHANGE DIAGRAM.

Suppose we are given morphisms $X_1, X_2 \rightarrow Y$ and $Y \rightarrow Z$. Show that the following diagram is a Cartesian square.

$$\begin{array}{ccc} X_1 \times_Y X_2 & \longrightarrow & X_1 \times_Z X_2 \\ \downarrow & & \downarrow \\ Y & \longrightarrow & Y \times_Z Y \end{array}$$

Assume all relevant (fibered) products exist. (If this exercise is too hard now, you can try it again at Exercise 1.3.B.)

Proof. ($Y \longrightarrow Y \times_Z Y$ is defined by identical embedding)

$$\begin{array}{ccc} X_1 \times_Y X_2 & \longrightarrow & X_2 \\ \downarrow & & \downarrow \\ X_1 & \longrightarrow & Y \\ & & \searrow \\ & & Z \end{array}$$

We need to prove the diagram below is a Cartesian Square:

$$\begin{array}{ccc} X_1 \times_Y X_2 & \longrightarrow & X_1 \times_Z X_2 \\ \downarrow & & \downarrow \\ Y & \longrightarrow & Y \times_Z Y \end{array}$$

Which becomes trivial when we notice

$$\begin{array}{ccccccc} X_1 \times_Y X_2 & & & & & & \\ & \searrow^{\exists!} & & \searrow^{\exists!} & & & \\ & X_1 \times_Z X_2 & \longrightarrow & X_2 & & & \\ & \downarrow & & \downarrow & & & \\ & X_1 & & Y \times_Z Y & \longrightarrow & Y & \\ & & & \downarrow & & \downarrow & \\ & & & Y & \longrightarrow & Z & \end{array}$$

$\square$

Exercise 2.T

Show that coproduct for Sets is disjoint union. This is why we use the notation $\coprod$ for disjoint union.

Proof. For

$$\begin{array}{ccccc} & & S & & \\ \psi_1 \nearrow & & \uparrow & & \nwarrow \psi_2 \\ S_1 & \xrightarrow{\varphi_1} & S_1 \sqcup S_2 & \xleftarrow{\varphi_2} & S_2 \end{array} .$$

For $s_1 \in S_1$, $s_2 \in S_2$, define

$$\varphi_1(s_1) = (s_1, \cdot), \quad \varphi_2(s_2) = (\cdot, s_2).$$

Now for any ψ_1, ψ_2 , we naturally have the induced morphism

$$\Psi : S_1 \sqcup S_2 \longrightarrow S$$

$$\Psi : (s_1, \cdot) \longmapsto \psi_1(s_1)$$

$$\Psi : (\cdot, s_2) \longmapsto \psi_2(s_2).$$

Which is unique up to isomorphism while making the diagram commutes. $\square$

Exercise 2.U

Suppose $A \rightarrow B$ and $A \rightarrow C$ are two ring morphisms, so in particular B and C are A -modules. Recall (Exercise 1.2.K) that $B \otimes_A C$ has a ring structure. Show that there is a natural morphism $B \rightarrow B \otimes_A C$ given by $b \mapsto b \otimes 1$. (This is not necessarily an inclusion; see Exercise 1.2.G.) Similarly, there is a natural morphism $C \rightarrow B \otimes_A C$. Show that this gives a fibered coproduct on rings, i.e., that

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ C & \longrightarrow & B \otimes_A C \end{array}$$

satisfies the universal property of fibered coproduct.

Proof. Denote

$$\begin{array}{ccccc} & & D & & \\ & \swarrow \varphi & \nearrow h & & \\ & B \otimes_A C & \longleftarrow & C & \\ & \uparrow & & \uparrow & \\ B & \longleftarrow & A & & \end{array}$$

and we obtain the unique map φ by

$$\varphi((a \cdot b) \otimes c) = g(b)$$

$$\varphi(b \otimes (a \cdot c)) = h(c)$$

$\square$

Exercise 2.V

Show that the composition of two monomorphisms is a monomorphism.

Proof.

$$\begin{array}{ccccc} & & A & & \\ & \swarrow \leq 1 & \searrow \leq 1 & & \\ X & \xrightarrow{\alpha} & Y & \xrightarrow{\beta} & Z \end{array}$$

From the graph we obtain the proof easily. $\square$

Exercise 2.W

Prove that a morphism $\pi : X \rightarrow Y$ is a monomorphism if and only if the fibered product $X \times_Y X$ exists, and the induced diagonal morphism $\delta_\pi : X \rightarrow X \times_Y X$ (Definition 1.2.7) is an isomorphism. We may then take this as the definition of monomorphism. (Monomorphisms aren't central to future discussions, although they will come up again. This exercise is just good practice.)

Proof. • When π is a monomorphism: For

$$\begin{array}{ccc} X \times_Y X & \longrightarrow & X \\ \downarrow & & \downarrow \pi \\ X & \xrightarrow{\pi} & Y \end{array}$$

From the definition of monomorphism, we know there's only one morphism in $\text{Mor}(X \times_Y X, X)$ making the diagram commutes. Therefore the fiber product must exist, where the morphism in between is an isomorphism.

- When $X \times_Y X$ exists, then any A letting the diagram below commute

$$\begin{array}{ccccc} A & & & & \\ & \searrow & & \searrow & \\ & X \times_Y X & \longrightarrow & X & \\ & \downarrow & & \downarrow \pi & \\ & X & \xrightarrow{\pi} & Y & \end{array}$$

whose morphism to $X \times_Y X$ will factor through $X \times_Y X$ uniquely. Therefore any two map:

$$\alpha, \beta : A \longrightarrow X$$

can be factorized into

$$\alpha = \alpha' \circ p, \beta = \beta' \circ p$$

Where $p : A \longrightarrow X \times_Y X$ and $\alpha', \beta' : X \times_Y X \longrightarrow X$.

$$\begin{array}{ccccc} A & & & & \\ & \searrow p & & \searrow \beta & \\ & X \times_Y X & \longrightarrow & X & \\ & \downarrow \alpha' & & \downarrow \pi & \\ & X & \xrightarrow{\pi} & Y & \end{array}$$

So if

$$\pi \circ \alpha = \pi \circ \beta$$

We'll have

$$\pi \circ \alpha' \circ p = \pi \circ \beta' \circ p$$

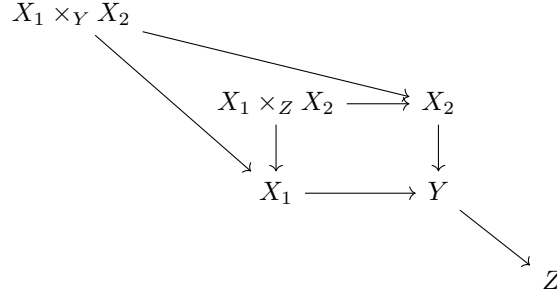
By the definition of diagonal morphism we know $\alpha' = \beta'$, therefore $\alpha' \circ p = \beta' \circ p$, i.e. $\alpha = \beta$. π satisfy the left cancellation law, so it's a monomorphism. □

Exercise 2.X

EASY EXERCISE.

We use the notation of Exercise 1.2.R. Show that if $Y \rightarrow Z$ is a monomorphism, then the morphism $X_1 \times_Y X_2 \rightarrow X_1 \times_Z X_2$ you described in Exercise 1.2.R is an isomorphism. (Hint: for any object V , give a natural bijection between maps from V to the first and maps from V to the second. It is also possible to use the Diagonal-Base-Change diagram, Exercise 1.2.S.)

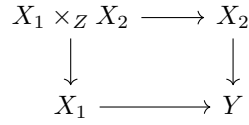
Proof.



From Exercise 1.2.R we know there exist a unique morphism

$$\varphi : X_1 \times_Y X_2 \longrightarrow X_1 \times_Z X_2.$$

Now since $Y \longrightarrow Z$ is monomorphism, from the commutativity of $X_1 \times_Z X_2 \rightarrow X_1 \rightarrow Y \rightarrow Z$ and $X_1 \times_Y X_2 \rightarrow X_1 \times_Z X_2 \rightarrow Y \rightarrow Z$ we can get the commutativity of



And therefore there's a unique morphism $\psi : X_1 \times_Z X_2 \longrightarrow X_1 \times_Y X_2$. By their uniqueness, we have

$$\varphi \circ \psi \circ \varphi = \varphi, \quad \psi \circ \varphi \circ \psi = \psi$$

So $\varphi \circ \psi = \text{Id}_{X_1 \times_Z X_2}$, $\psi \circ \varphi = \text{Id}_{X_1 \times_Y X_2}$. □

Exercise 2.Y

IMPORTANT EXERCISE THAT YOU SHOULD DO ONCE IN YOUR LIFE (YONEDA'S LEMMA).

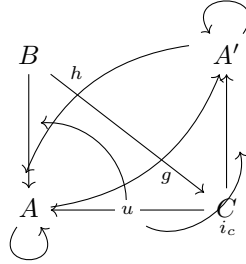
(a) Suppose you have two objects A and A' in a category $\mathcal{C}$, and maps

$$i_C : \text{Mor}(C, A) \longrightarrow \text{Mor}(C, A')$$

that commute with the maps (1.2.11.1). Show that the i_C (as C ranges over the objects of $\mathcal{C}$) are induced from a unique morphism $g : A \rightarrow A'$. More precisely, show that there is a unique morphism $g : A \rightarrow A'$ such that for all $C \in \mathcal{C}$, i_C is $u \mapsto g \circ u$.

(b) If furthermore the i_C are all bijections, show that the resulting g is an isomorphism. (Hint for both: This is much easier than it looks. This statement is so general that there are really only a couple of things that you could possibly try. For example, if you're hoping to find a morphism $A \rightarrow A'$, where will you find it? Well, you are looking for an element $\text{Mor}(A, A')$. So just plug in $C = A$ to (1.2.11.2), and see where the identity goes.)

Proof.



1. We simply let Id_C correspond to the morphism $A \longrightarrow A'$ and verify the properties stated in the problems. Also, uniqueness can be obtained by letting $C = A$.

2. We follow from the hint: Let $C = A'$ and we get a bijection

$$\text{Id}_{A'} : (\text{Mor}(A', A)) \longrightarrow (\text{Mor}(A', A'))$$

We pick out h from $\text{Mor}(A', A)$ such that $\text{Id}_{A'}(h) = \text{Id}_{A'}$. From (a) we know $\text{Id}_{A'}$ can be represented using g :

$$\text{Id}_{A'} = \text{Id}_{A'}(h) = g \circ h$$

On the other hand, we prove Id_A^{-1} can be induced by h using the similar method to (a).

$$\exists! \text{Id}_A \in \text{Mor}(A, A), \text{Id}_A(\text{Id}_A) = g \in \text{Mor}(A, A)$$

And we can see

$$\text{Id}_A = \text{Id}^{-1}(g) = h \circ g$$

□

Exercise 2.Z [Skipped temporarily due to the star.]

*EXERCISE.

- (a) Suppose A and B are objects in a category $\mathcal{C}$. Give a bijection between the natural transformations $h^A \rightarrow h^B$ of covariant functors $\mathcal{C} \rightarrow \text{Sets}$ (see Example 1.1.14 for the definition) and the morphisms $B \rightarrow A$.
- (b) State and prove the corresponding fact for contravariant functors h_A (see Example 1.1.20). Remark: A contravariant functor F from $\mathcal{C}$ to Sets is said to be representable if there is a natural isomorphism

$$\xi : F \xrightarrow{\sim} h_A.$$

Thus the representing object A is determined up to unique isomorphism by the pair (F, ξ) . There is a similar definition for covariant functors. (We will revisit this in §7.6, and this problem will appear again as Exercise 7.6.C. The element $\xi^{-1}(\text{id}_A) \in F(A)$ is often called the “universal object”; do you see why?)

- (c) Yoneda’s Lemma. Suppose F is a covariant functor $\mathcal{C} \rightarrow \text{Sets}$, and $A \in \mathcal{C}$. Give a bijection between the natural transformations $h^A \rightarrow F$ and $F(A)$. (The corresponding fact for contravariant functors is essentially Exercise 10.1.B.)

Proof. 1. We consider the contravariant functors of the morphisms $B \rightarrow A$.

2.

□

3 Limits and colimits

Exercise 3.A [Trivial]

EXERCISE (REALITY CHECK).

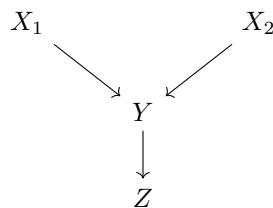
Suppose that the partially ordered set I has an initial object e . Show that the limit of any diagram indexed by I exists.

Proof. Since $\mathcal{I}$ is a partially ordered set, there’s at most one morphism for any two object in $\mathcal{I}$. □

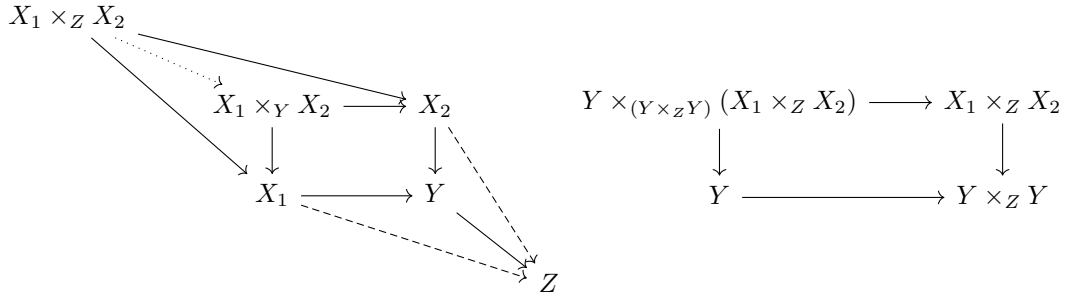
Exercise 3.B

EXERCISE: THE DIAGONAL-BASE-CHANGE DIAGRAM, AGAIN.

Solve 1.2.S again by identifying both $X_1 \times_Y X_2$ and $Y \times_{(Y \times_Z Y)} (X_1 \times_Z X_2)$ as the limit of the diagram



Proof.



For $X_1 \times_Y X_2$:

$$\begin{aligned} X_1 \times_Y X_2 \rightarrow Y &= X_1 \times_Y X_2 \rightarrow X_1 \rightarrow Y \\ &= X_1 \times_Y X_2 \rightarrow X_2 \rightarrow Y \\ X_1 \times_Y X_2 \rightarrow Z &= X_1 \times_Y X_2 \rightarrow Y \rightarrow Z \end{aligned}$$

For $Y \times_{(Y \times_Z Y)} (X_1 \times_Z X_2)$:

$$\begin{aligned} Y \times_{(Y \times_Z Y)} (X_1 \times_Z X_2) \rightarrow Y &= Y \times_{(Y \times_Z Y)} (X_1 \times_Z X_2) \rightarrow X_1 \times_Z X_2 \rightarrow X_1 \rightarrow Y \\ &= Y \times_{(Y \times_Z Y)} (X_1 \times_Z X_2) \rightarrow X_1 \rightarrow Y \\ Y \times_{(Y \times_Z Y)} (X_1 \times_Z X_2) \rightarrow Z &= Y \times_{(Y \times_Z Y)} (X_1 \times_Z X_2) \rightarrow Y \rightarrow Z \end{aligned}$$

□

Exercise 3.C

IMPORTANT EXERCISE.

Show that in the category *Sets*,

$$\left\{ (a_i)_{i \in \mathcal{I}} \in \prod_i A_i : F(m)(a_j) = a_k \text{ for all } m \in \text{Mor}_{\mathcal{I}}(j, k) \subset \text{Mor}(\mathcal{I}) \right\},$$

along with the obvious projection maps to each A_i , is the limit $\text{Lim}_{\mathcal{I}} A_i$.

This clearly also works in the category Mod_A of A -modules (in particular Vec_k and Ab), as well as *Rings*.

Proof.

$$\begin{array}{ccc} (a_i)_{i \in \mathcal{I}} & & \\ \downarrow & \searrow & \\ A_j & \longrightarrow & A_k \end{array}$$

Because from the definition of $(a_i)_{i \in \mathcal{I}}$, we have

$$F(m)(a_j) = a_k$$

So $(a_i)_{i \in \mathcal{I}} \rightarrow A_j \rightarrow A_k$ commutes with $(a_i)_{i \in \mathcal{I}} \rightarrow A_k$. □

Exercise 3.D (a) Interpret the statement “ $\mathbb{Q} = \text{colim} \frac{1}{n} \mathbb{Z}$ ”.

(b) Interpret the union of some subsets of a given set as a colimit. (Dually, the intersection can be interpreted as a limit.) The objects of the category in question are the subsets of the given set.

Solution. 1. It means fractions converting into different denominators stands for the same rational number, which is obviously true.

2. Because we can view the set category as posets, where morphisms in between two objects are unique.

Exercise 3.E

Suppose $\mathcal{I}$ is filtered. (We will almost exclusively use the case where $\mathcal{I}$ is a filtered set.) Recall the symbol $\coprod$ for disjoint union of sets. Show that any diagram in Sets indexed by $\mathcal{I}$ has the following, with the obvious maps to it, as a colimit:

$$\left\{ (a_i, i) \in \coprod_{i \in \mathcal{I}} A_i \right\} / \{ (a_i, i) \sim (a_j, j) \text{ if and only if there are } f: A_i \rightarrow A_k \text{ and } g: A_j \rightarrow A_k \text{ in the diagram for which } f(a_i) = g(a_j) \}$$

(You will see that the “filtered” hypothesis is there to ensure that $\sim$ is an equivalence relation.)

Proof. The proof is done by definition. □

Exercise 3.F

Verify that the A -module described above is indeed the colimit. (Make sure you verify that addition is well-defined, i.e., is independent of the choice of representatives m_i and m_j , the choice of ℓ , and the choice of arrows u and v . Similarly, make sure that scalar multiplication is well-defined.)

Proof. (Trivial, skipped.) □

Exercise 3.G

USEFUL EXERCISE (LOCALIZATION AS A COLIMIT).

Generalize Exercise 1.3.D(a) to interpret localization of an integral domain as a colimit over a filtered set: suppose S is a multiplicative set of A , and interpret $S^{-1}A = \text{colim}_s \frac{1}{s}A$ where the colimit is over $s \in S$, and in the category of A -modules. (Aside: Can you make some version of this work even if A isn't an integral domain, e.g., $S^{-1}A = \text{colim} A_s$? This will work in the category of A -algebras.)

Proof. (The proof is similar to 1.3.D I'll skip.) □

Exercise 3.H

EXERCISE: COLIMITS OF A -MODULES WITHOUT THE FILTERED CONDITION.

Suppose you are given a diagram of A -modules indexed by $\mathcal{J}$: $F: \mathcal{J} \rightarrow \text{Mod}_A$, where we let $M_i := F(i)$. Show that the colimit is $\oplus_{i \in \mathcal{J}} M_i$ modulo the relations $m_i - F(n)(m_i)$ for every $n: i \rightarrow j$ in $\mathcal{J}$ (i.e., for every arrow in the diagram). (Somewhat more precisely: “modulo” means “quotiented by the submodule generated by”.)

Proof. (Trivial, skipped.)

$$\begin{array}{ccc} \text{Colim}_{i \in \mathcal{J}} M_i & & \\ \uparrow & \nwarrow & \\ M_i & \longleftarrow & M_j \end{array}$$

□

4 Adjoints

Exercise 4.A

Write down what this diagram should be.

Solution.

$$\begin{array}{ccc} \text{Mor}_{\mathcal{B}}(F(A), B) & \xrightarrow{g^*} & \text{Mor}_{\mathcal{B}}(F(A), B') \\ \downarrow \tau_{AB} & & \downarrow \tau_{AB'} \\ \text{Mor}_{\mathcal{A}}(A, G(B)) & \xrightarrow{Gg^*} & \text{Mor}_{\mathcal{A}}(A, G(B')) \end{array}$$

Exercise 4.B

Show that the map τ_{AB} (1.4.0.1) has the following properties. For each A there is a map $\eta_A : A \rightarrow GF(A)$ so that for any $g : F(A) \rightarrow B$, the corresponding $\tau_{AB}(g) : A \rightarrow G(B)$ is given by the composition

$$A \xrightarrow{\eta_A} GF(A) \xrightarrow{Gg} G(B).$$

Similarly, there is a map $\epsilon_B : FG(B) \rightarrow B$ for each B so that for any $f : A \rightarrow G(B)$, the corresponding map $\tau_{AB}^{-1}(f) : F(A) \rightarrow B$ is given by the composition

$$F(A) \xrightarrow{Ff} FG(B) \xrightarrow{\epsilon_B} B.$$

Proof. Use the natural property: For all $g : B' \rightarrow B$, we have

$$\begin{array}{ccc} \text{Mor}_{\mathcal{B}}(F(A), B') & \xrightarrow{g^*} & \text{Mor}_{\mathcal{B}}(F(A), B) \\ \downarrow \tau_{AB'} & & \downarrow \tau_{AB} \\ \text{Mor}_{\mathcal{A}}(A, G(B')) & \xrightarrow{Gg^*} & \text{Mor}_{\mathcal{A}}(A, G(B)) \end{array}$$

commutes. Therefore let $B' = F(A)$ and we'll have

$$\begin{array}{ccc} \text{Mor}_{\mathcal{B}}(F(A), F(A)) & \xrightarrow{g^*} & \text{Mor}_{\mathcal{B}}(F(A), B) \\ \downarrow \tau_{AF(A)} & & \downarrow \tau_{AB} \\ \text{Mor}_{\mathcal{A}}(A, GF(A)) & \xrightarrow{Gg^*} & \text{Mor}_{\mathcal{A}}(A, G(B)) \end{array}$$

commutes. Take $\text{Id}_{F(A)} \in \text{Mor}_{\mathcal{B}}(F(A), F(A))$ we would have

$$\begin{aligned} \tau_{AB}(g) &= \tau_{AB} \circ g \circ \text{Id}_{F(A)} \\ &= G(g) \circ \tau_{AF(A)} \circ \text{Id}_{F(A)}. \end{aligned}$$

Let $\eta_A = \tau_{AF(A)}(\text{Id}_{F(A)})$ and we're done.

On the other hand: vice versa. □

Exercise 4.C

Suppose M , N , and P are A -modules (where A is a ring). Describe a bijection $\text{Hom}_A(M \otimes_A N, P) \leftrightarrow \text{Hom}_A(M, \text{Hom}_A(N, P))$. (Hint: try to use the universal property of $\otimes$.)

Proof. For $\forall \beta \in \text{Bil}(M, N; P)$, we may define

$$f : M \longrightarrow \text{Hom}_A(N, P)$$

$$m \longmapsto \beta(m, (-))$$

i.e. $f(m)(n) = \beta(m, n)$.

For $\forall f \in \text{Hom}_A(M, \text{Hom}_A(N, P))$, let

$$\beta : M \times N \longrightarrow P$$

$$(m, n) \longmapsto f(m)(n)$$

i.e. $\beta(m, n) = f(m)(n)$.

This defines the bijection. □

Exercise 4.D

EXERCISE (TENSOR-HOM ADJUNCTION).

Show that $(\cdot) \otimes_A N$ and $\text{Hom}_A(N, \cdot)$ are adjoint functors.

Proof. Define F, G to be the functor $(\cdot) \otimes_A N, \text{Hom}_A(N, \cdot)$ respectively. We need to prove those τ_{XY} are natural, i.e. for $\forall \varphi : M \longrightarrow M', \psi : P \longrightarrow P'$

$$\begin{array}{ccc} \text{Hom}_A(M' \otimes_A N, P) & \xrightarrow{F\varphi^*} & \text{Hom}_A(M \otimes_A N, P) \\ \downarrow \tau_{M'P} & & \downarrow \tau_{MP} \\ \text{Hom}_A(M', \text{Hom}_A(N, P)) & \xrightarrow{\varphi^*} & \text{Hom}_A(M, \text{Hom}_A(N, P)) \end{array}$$

and

$$\begin{array}{ccc} \text{Hom}_A(M \otimes_A N, P) & \xrightarrow{\psi^*} & \text{Hom}_A(M \otimes_A N, P') \\ \downarrow \tau_{MP} & & \downarrow \tau_{MP'} \\ \text{Hom}_A(M, \text{Hom}_A(N, P)) & \xrightarrow{G\psi^*} & \text{Hom}_A(M, \text{Hom}_A(N, P')) \end{array}$$

commutes. This is obvious when we consider the property of homomorphisms. $\square$

Exercise 4.E

Suppose $B \rightarrow A$ is a morphism of rings. If M is an A -module, you can create a B -module M_B by considering it as a B -module. This gives a functor $\cdot_B : \text{Mod}_A \rightarrow \text{Mod}_B$. Show that this functor is right-adjoint to $\cdot \otimes_B A$. In other words, describe a bijection

$$\text{Hom}_A(N \otimes_B A, M) \cong \text{Hom}_B(N, M_B)$$

functorial in both arguments. (This adjoint pair is very important.)

Proof. (I) τ_{NM} : For $\Phi : N \otimes_B A \longrightarrow M$, define

$$\begin{aligned} \phi : N &\longrightarrow M_B \\ n &\longmapsto \Phi(n \otimes_B 1_A) \end{aligned}$$

τ_{NM}^{-1} : On the other hand, for $\phi : N \longrightarrow M_B$, define

$$\Phi : N \otimes_B A \longrightarrow M$$

by

$$\begin{array}{ccc} N \times A & \xrightarrow{\phi \times \text{Id}} & M_B \times A \\ \downarrow \pi & & \searrow \\ N \times_B A & \xrightarrow{\Phi} & M \end{array}$$

and we have

$$\Phi : n \otimes_B a \longmapsto a\phi(n)$$

Obviously the maps defined above are inverses to each other.

(II) We need to show such bijection (transformation) is natural, i.e. $\forall f \in \text{Hom}_A(M, M')$,

$$\begin{array}{ccc} \text{Hom}_A(N \otimes_B A, M) & \xrightarrow{f^*} & \text{Hom}_A(N \otimes_B A, M') \\ \downarrow \tau_{NM} & & \downarrow \tau_{NM'} \\ \text{Hom}_B(N, M_B) & \xrightarrow{(f^*)_B} & \text{Hom}_B(N, M'_B) \end{array}$$

commute, while $\forall g \in \text{Hom}_B(N, N')$,

$$\begin{array}{ccc} \text{Hom}_A(N' \otimes_B A, M) & \xrightarrow{g^* \otimes_B \text{Id}_A} & \text{Hom}_A(N \otimes_B A, M) \\ \downarrow \tau_{N'M} & & \downarrow \tau_{NM} \\ \text{Hom}_B(N', M_B) & \xrightarrow{g^*} & \text{Hom}_B(N, M_B) \end{array}$$

commutes.

Pick a homomorphism from the start, then track it through the diagram directly and we're done. $\square$

Exercise 4.F [Trivial]

EXERCISE (AN ABELIAN GROUP IS GROUPIFIED BY ITSELF).

Show that if an abelian semigroup is already a group then the identity morphism is the groupification. (More correct: the identity morphism is a groupification.) Note that you don't need to construct groupification (or even know that it exists in general) to solve this exercise.

Exercise 4.G [Follow the hint, trivial]

Construct the "groupification functor" H from the category of nonempty abelian semigroups to the category of abelian groups. (One possible construction: given an abelian semigroup S , the elements of its groupification $H(S)$ are ordered pairs $(a, b) \in S \times S$, which you may think of as $a - b$, with the equivalence that $(a, b) \sim (c, d)$ if $a + d + e = b + c + e$ for some $e \in S$. Describe addition in this group, and show that it satisfies the properties of an abelian group. Describe the abelian semigroup map $S \rightarrow H(S)$.) Let F be the forgetful functor from the category of abelian groups $\mathbf{Ab}$ to the category of abelian semigroups. Show that H is left-adjoint to F .

Solution. (It's just like constructing a fraction field of a ring.)

Exercise 4.H

EXERCISE (CF. EXERCISE 1.4.E).

The purpose of this exercise is to give you more practice with "left adjoints of forgetful functors", the means by which we get abelian groups from abelian semigroups, and sheaves from presheaves. Suppose A is a ring, and S is a multiplicative subset. Then $S^{-1}A$ -modules are a full subcategory (§1.1.15) of the category of A -modules (via the obvious inclusion $\text{Mod}_{S^{-1}A} \hookrightarrow \text{Mod}_A$). Then $\text{Mod}_A \rightarrow \text{Mod}_{S^{-1}A}$ can be interpreted as an adjoint to the forgetful functor $\text{Mod}_{S^{-1}A} \rightarrow \text{Mod}_A$. State and prove the correct statements.

(Here is the larger story. Every $S^{-1}A$ -module is an A -module, and this is an injective map, so we have a forgetful functor $F : \text{Mod}_{S^{-1}A} \rightarrow \text{Mod}_A$. In fact this is a fully faithful functor: it is injective on objects, and the morphisms between any two $S^{-1}A$ -modules as A -modules are just the same when they are considered as $S^{-1}A$ -modules. Then there is a functor $G : \text{Mod}_A \rightarrow \text{Mod}_{S^{-1}A}$, which might reasonably be called "localization with respect to S ", which is left-adjoint to the forgetful functor. Translation: If M is an A -module, and N is an $S^{-1}A$ -module, then $\text{Mor}(GM, N)$ (morphisms as $S^{-1}A$ -modules, which are the same as morphisms as A -modules) are in natural bijection with $\text{Mor}(M, FN)$ (morphisms as A -modules).)

Proof. From **Exercise 1.4.E**, for ring homomorphism

$$A \longrightarrow S^{-1}A$$

we have

$$\text{Hom}_{S^{-1}A}(S^{-1}N, M) \cong \text{Hom}_A(N, M_A)$$

(I suppose the rest is trivial.) $\square$

5 An introduction to abelian categories

Exercise 5.A [Trivial]

Describe exact sequences

$$\begin{aligned} 0 &\longrightarrow \text{Im } f^i \longrightarrow A^{i+1} \longrightarrow \text{Coker } f^i \longrightarrow 0 \\ 0 &\longrightarrow H^i(A^\bullet) \longrightarrow \text{Coker } f^{i-1} \longrightarrow \text{Im } f^i \longrightarrow 0 \end{aligned}$$

(These are somehow dual to (1.5.6.3). In fact in some mirror universe this might have been given as the standard definition of homology.) Assume the category is that of modules over a fixed ring for convenience, but be aware that the result is true for any abelian category.

Exercise 5.B*EXERCISE AND IMPORTANT DEFINITION.**Suppose*

$$0 \xrightarrow{d^0} A^1 \xrightarrow{d^1} \dots \xrightarrow{d^{n-1}} A^n \xrightarrow{d^n} 0$$

is a complex of finite-dimensional k -vector spaces (often called $A^\bullet$ for short). Define $h^i(A^\bullet) := \dim H^i(A^\bullet)$. Show that $\sum (-1)^i \dim A^i = \sum (-1)^i h^i(A^\bullet)$. In particular, if $A^\bullet$ is exact, then $\sum (-1)^i \dim A^i = 0$. (If you haven't dealt much with cohomology, this will give you some practice.)

Proof.

$$\begin{aligned} \sum_{i=1}^n (-1)^i \dim A^i &= \sum_{i=1}^n (-1)^i (\dim \ker f^i + \dim \operatorname{Im} f^i) \\ &= \sum_{i=1}^n (-1)^i (\dim \ker f^i - \dim \operatorname{Im} f^{i-1}) \\ &= \sum_{i=1}^n (-1)^i h^i(A^\bullet) \end{aligned}$$

□

Exercise 5.C*IMPORTANT EXERCISE.*

Suppose $\mathcal{C}$ is an abelian category. Define the category $\operatorname{Com}_{\mathcal{C}}$ of complexes as follows. The objects are infinite complexes

$$A^\bullet: \dots \longrightarrow A^{i-1} \xrightarrow{f^{i-1}} A^i \xrightarrow{f^i} A^{i+1} \xrightarrow{f^{i+1}} \dots$$

in $\mathcal{C}$, and the morphisms $A^\bullet \rightarrow B^\bullet$ are commuting diagrams

$$\begin{array}{ccccccc} \dots & \longrightarrow & A^i & \xrightarrow{f^i} & A^{i+1} & \longrightarrow & \dots \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \dots & \longrightarrow & B^i & \xrightarrow{g^i} & B^{i+1} & \longrightarrow & \dots \end{array}$$

Show that $\operatorname{Com}_{\mathcal{C}}$ is an abelian category.

Feel free to deal with the special case of modules over a fixed ring. (Remark for experts: Essentially the same argument shows that $\mathcal{C}^{\mathcal{I}}$ is an abelian category for any small category $\mathcal{I}$ and any abelian category $\mathcal{C}$. This immediately implies that the category of presheaves on a topological space X with values in an abelian category $\mathcal{C}$ is automatically an abelian category, cf. §2.3.5.)

Proof. This is obtained immediately from $\mathcal{C}$ being abelian category. □

Exercise 5.D*IMPORTANT EXERCISE.*

Show that (1.5.6.5) induces a map of homology $H^i(A^\bullet) \rightarrow H^i(B^\bullet)$. Show furthermore that H^i is a covariant functor $\operatorname{Com}_{\mathcal{C}} \rightarrow \mathcal{C}$. (Again, feel free to deal with the special case Mod_A .)

Proof. (i) By (1.5.6.3) we know any chain complex

$$\dots \longrightarrow A^{i-1} \xrightarrow{f^{i-1}} A^i \xrightarrow{f^i} A^{i+1} \xrightarrow{f^{i+1}} \dots$$

can be decomposed into series of

$$0 \longrightarrow \ker f^i \longrightarrow A^i \longrightarrow \operatorname{Im} f^i \longrightarrow 0$$

and

$$0 \longrightarrow \operatorname{Im} f^{i-1} \longrightarrow \ker f^i \longrightarrow H^i(A^\bullet) \longrightarrow 0.$$

Therefore the any homomorphism bewteen chain complexes naturally induces homomorphisms such as

$$\text{Ker } f^i \longrightarrow \text{Ker } g^i$$

and

$$\text{Im } f^i \longrightarrow \text{Im } g^i.$$

Hence

$$\text{Ker } f^i / \text{Im } f^{i-1} \longrightarrow \text{Ker } g^i / \text{Im } g^{i-1}$$

i.e.

$$H^i(A^\bullet) \longrightarrow H^i(B^\bullet).$$

Done.

(ii) (*I think it's trivial.*)

□

Exercise 5.E

Show that two homotopic maps give the same map on homology.

Proof.

$$\begin{array}{ccccccc} \dots & \longrightarrow & C^{i-1} & \xrightarrow{d_C^{i-1}} & C^i & \xrightarrow{d_C^i} & C^{i+1} & \xrightarrow{d_C^{i+1}} & \dots \\ & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \dots & \longrightarrow & D^{i-1} & \xrightarrow{d_D^{i-1}} & D^i & \xrightarrow{d_D^i} & D^{i+1} & \xrightarrow{d_D^{i+1}} & \dots \end{array}$$

This is euqivalent to prove $f - g$ is gives the zero mapping on homology, i.e. $dw - wd$ gives the zero mapping on homology, which is trivial. □

Exercise 5.F [Trivial]

Suppose F is an exact functor. Show that applying F to an exact sequence preserves exactness. For example, if F is covariant, and $A' \rightarrow A \rightarrow A''$ is exact, then $FA' \rightarrow FA \rightarrow FA''$ is exact. (This will be generalized in Exercise 1.5.I(c).)

Proof. (Just decompose those sequences)

□

Exercise 5.G

Suppose A is a ring, $S \subset A$ is a multiplicative subset, and M is an A -module.

- Show that localization of A -modules $\text{Mod}_A \rightarrow \text{Mod}_{S^{-1}A}$ is an exact covariant functor.
- Show that $(\cdot) \otimes_A M$ is a right-exact covariant functor $\text{Mod}_A \rightarrow \text{Mod}_A$. (This is a repeat of Exercise 1.2.H.)
- Show that $\text{Hom}(M, \cdot)$ is a left-exact covariant functor $\text{Mod}_A \rightarrow \text{Mod}_A$. If $\mathcal{C}$ is any abelian category, and $C \in \mathcal{C}$, show that $\text{Hom}(C, \cdot)$ is a left-exact covariant functor $\mathcal{C} \rightarrow \text{Ab}$.
- Show that $\text{Hom}(\cdot, M)$ is a left-exact contravariant functor $\text{Mod}_A \rightarrow \text{Mod}_A$. If $\mathcal{C}$ is any abelian category, and $C \in \mathcal{C}$, show that $\text{Hom}(\cdot, C)$ is a left-exact contravariant functor $\mathcal{C} \rightarrow \text{Ab}$.

Proof. (a) Trivial.

- (i) See [Introduction to Commutative Algebra, **Proposition 2.18** – M. Atiyah], trivial.
- (ii)
- (i) See [Introduction to Commutative Algebra, **Proposition 2.9** – M. Atiyah], trivial.
- (ii)
- (i) See [Introduction to Commutative Algebra, **Proposition 2.9** – M. Atiyah], trivial.
- (ii)

□

Exercise 5.H

Suppose M is a finitely presented A -module: M has a finite number of generators, and with these generators it has a finite number of relations; or usefully equivalently, fits in an exact sequence

$$A^{\oplus q} \longrightarrow A^{\oplus p} \longrightarrow M \longrightarrow 0.$$

Use (1.5.9.1) and the left-exactness of Hom to describe an isomorphism

$$S^{-1} \text{Hom}_A(M, N) \xrightarrow{\sim} \text{Hom}_{S^{-1}A}(S^{-1}M, S^{-1}N).$$

(You might be able to interpret this in light of a variant of Exercise 1.5.I below, for left-exact contravariant functors rather than right-exact covariant functors.)

Proof. First, by

$$A^{\oplus q} \longrightarrow A^{\oplus p} \longrightarrow A^M \longrightarrow 0$$

being exact, we have

$$S^{-1}A^{\oplus q} \longrightarrow S^{-1}A^{\oplus p} \longrightarrow S^{-1}A^M \longrightarrow 0$$

being exact, hence

$$0 \longrightarrow \text{Hom}_{S^{-1}A}(S^{-1}A^{\oplus q}, S^{-1}N) \longrightarrow \text{Hom}_{S^{-1}A}(S^{-1}A^{\oplus p}, S^{-1}N) \longrightarrow \text{Hom}_{S^{-1}A}(S^{-1}A^M, S^{-1}N)$$

is exact.

On the other hand,

$$A^{\oplus q} \longrightarrow A^{\oplus p} \longrightarrow A^M \longrightarrow 0$$

being exact implies

$$0 \longrightarrow \text{Hom}_A(A^{\oplus q}, N) \longrightarrow \text{Hom}_A(A^{\oplus p}, N) \longrightarrow \text{Hom}_A(A^M, N)$$

is exact, therefore

$$0 \longrightarrow S^{-1} \text{Hom}_A(A^{\oplus q}, N) \longrightarrow S^{-1} \text{Hom}_A(A^{\oplus p}, N) \longrightarrow S^{-1} \text{Hom}_A(A^M, N)$$

is exact.

It's easy to define a natural homomorphism $S^{-1} \text{Hom}_A(M, N) \longrightarrow \text{Hom}_{S^{-1}A}(S^{-1}M, S^{-1}N)$ explicitly. Also, obviously

$$\text{Hom}_{S^{-1}A}(S^{-1}A, S^{-1}N) \cong S^{-1}N \text{ by } \bar{f} \mapsto f(1)$$

and

$$S^{-1} \text{Hom}_A(M, N) \cong S^{-1}N \text{ by } \frac{f}{s} \mapsto \frac{f(1)}{s}.$$

Therefore we have the direct sum

$$\text{Hom}_{S^{-1}A}(S^{-1}A^{\oplus r}, S^{-1}N) \cong (S^{-1}N)^{\oplus r} \cong S^{-1} \text{Hom}_A(M^{\oplus r}, N).$$

□

Exercise 5.I [Beautiful!]

IMPORTANT EXERCISE (THE FHHF THEOREM).

This result can take you far, and perhaps for that reason it has sometimes been called the *Fernbahnhof (FernbaHnHoF) Theorem*, notably in [Vak1, Exer. 1.5.I]. Suppose $F : \mathcal{A} \rightarrow \mathcal{B}$ is a covariant functor of abelian categories, and $C^\bullet$ is a complex in $\mathcal{A}$.

- (a) (F right-exact yields $FH^\bullet \longrightarrow H^\bullet F$) If F is right-exact, describe a natural morphism $FH^\bullet \rightarrow H^\bullet F$. (More precisely, for each i , the left side is F applied to the cohomology at piece i of $C^\bullet$, while the right side is the cohomology at piece i of $FC^\bullet$.)
- (b) (F left-exact yields $FH^\bullet \longleftarrow H^\bullet F$) If F is left-exact, describe a natural morphism $H^\bullet F \rightarrow FH^\bullet$.
- (c) (F exact yields $FH^\bullet \xrightarrow{\sim} H^\bullet F$) If F is exact, show that the morphisms of (a) and (b) are inverses and thus isomorphisms.

Hint for (a): use $C^i \xrightarrow{d^i} C^{i+1} \rightarrow \text{coker } d^i \rightarrow 0$ to give an isomorphism $F \text{coker } d^i \xrightarrow{\sim} \text{coker } Fd^i$. Then use the first line of (1.5.6.4) to give an epimorphism $F \text{im } d^i \rightarrow \text{im } Fd^i$. Then use the second line of (1.5.6.4) to give the desired map $FH^i C^\bullet \rightarrow H^i FC^\bullet$. While you are at it, you may as well describe a map for the fourth member of the quartet $\text{coker}, \text{im}, H, \text{ker}$: $F \text{ker } d^i \rightarrow \text{ker } Fd^i$.

Proof. (Note that in homological algebra, it's a consequence that every left, right exact, or exact, functor being additive. Therefore it's well-defined to write $H^\bullet F$.)

(i) We follow the hint: consider the exact sequence

$$C^i \xrightarrow{d^i} C^{i+1} \rightarrow \text{Coker } d^i \rightarrow 0$$

and apply it with functor F , we have

$$F(C^i) \xrightarrow{F(d^i)} F(C^{i+1}) \rightarrow F \text{Coker } d^i \rightarrow 0$$

being exact. But at the same time we have the exact sequence

$$F(C^i) \xrightarrow{F(d^i)} F(C^{i+1}) \rightarrow \text{Coker } Fd^i \rightarrow 0.$$

Hence $F \text{Coker } d^i \cong \text{Coker } Fd^i$.

From (1.5.6.4), we know

$$0 \rightarrow \text{Im } d^i \rightarrow C^{i+1} \rightarrow \text{Coker } d^i \rightarrow 0$$

is exact. Apply it with F ,

$$F \text{Im } d^i \rightarrow F C^{i+1} \rightarrow F \text{Coker } d^i \rightarrow 0$$

is exact.

$$\begin{array}{ccccccc} & & & & \text{Coker } Fd^i & & \\ & & & & \uparrow & \downarrow & \\ F \text{Im } d^i & \xrightarrow{\quad} & F C^{i+1} & \xrightarrow{\quad} & F \text{Coker } d^i & \xrightarrow{\quad} & 0 \\ & \searrow \text{dotted} & \nearrow & & & & \\ & & \text{Im } Fd^i & & & & \end{array}$$

and we have the epimorphism

$$F \text{Im } d^i \rightarrow \text{Im } Fd^i \rightarrow 0$$

Again from (1.5.6.4), we know

$$0 \rightarrow H^i(C^\bullet) \rightarrow \text{Coker } d^{i-1} \rightarrow \text{Im } d^i \rightarrow 0$$

is exact. Apply it with F ,

$$\begin{array}{ccccccc} FH^i(C^\bullet) & \longrightarrow & F(\text{Coker } d^{i-1}) & \longrightarrow & F(\text{Im } d^i) & \longrightarrow & 0 \\ \downarrow & & \downarrow \sim & & \downarrow & & \\ 0 & \longrightarrow & H^i F(C^\bullet) & \longrightarrow & \text{Coker } Fd^{i-1} & \longrightarrow & \text{Im } Fd^i \longrightarrow 0 \end{array}$$

we obtain the natural morphism by universal property.

(ii) Similar to (i), we use the two short exact sequences in (1.5.6.3) and obtain

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Im } Fd^{i-1} & \longrightarrow & \text{Ker } Fd^i & \longrightarrow & H^i(FC^\bullet) \longrightarrow 0 \\ & & \downarrow & & \downarrow \sim & & \downarrow \\ 0 & \longrightarrow & F \text{Im } d^{i-1} & \longrightarrow & F \text{Ker } d^i & \longrightarrow & FH^i(C^\bullet) \end{array}.$$

Done

(iii) If F is exact, the epimorphism

$$F\text{Im } d^i \longrightarrow \text{Im } Fd^i \longrightarrow 0$$

obtained in (i) becomes isomorphism

$$F\text{Im } d^i \cong \text{Im } Fd^i,$$

therefore the induced natural morphism is an isomorphism. The same story goes to (ii). Thus the morphism in (i) and (ii), denoted by α, β , are both isomorphisms. Now we'll prove they're actually inverses to each other:

Consider

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Ker } F(d^i) & \longrightarrow & H^i F(C^\bullet) & \hookrightarrow & \text{Coker } F(d^{i-1}) \longrightarrow 0 \\ & & \downarrow \sim & & \downarrow \beta & & \downarrow \sim \\ 0 & \longrightarrow & F(\text{Ker } d^i) & \longrightarrow & F(H^i(C^\bullet)) & \hookrightarrow & F(\text{Coker } d^{i-1}) \longrightarrow 0 \\ & & \uparrow \sim & & \downarrow \alpha & & \downarrow \sim \\ 0 & \longrightarrow & \text{Ker } F(d^i) & \longrightarrow & H^i F(C^\bullet) & \hookrightarrow & \text{Coker } F(d^{i-1}) \longrightarrow 0 \end{array}$$

We have

$$(\text{mono}) \circ (\alpha \circ \beta) \circ (\text{epi}) = (\text{mono}) \circ \text{id}_{H^i F(C^\bullet)} \circ (\text{epi})$$

By the cancellation law of monomorphism and epimorphism, we have

$$\alpha \circ \beta = \text{Id}$$

□

Exercise 5.J

EXERCISE (Kernels COMMUTE WITH LIMITS).

Suppose $\mathcal{C}$ is an abelian category, and $a: \mathcal{I} \rightarrow \mathcal{C}$ and $b: \mathcal{I} \rightarrow \mathcal{C}$ are two diagrams in $\mathcal{C}$ indexed by $\mathcal{I}$. For convenience, let $A_i = a(i)$ and $B_i = b(i)$ be the objects in those two diagrams. Let $h_i: A_i \rightarrow B_i$ be maps commuting with the maps in the diagram. (Translation: h is a natural transformation of functors $a \rightarrow b$, see §1.1.21.) Then the $\text{Ker } h_i$ form another diagram in $\mathcal{C}$ indexed by $\mathcal{I}$. Describe a canonical isomorphism $\text{Lim Ker } h_i \xleftarrow{\sim} \text{Ker}(\text{Lim } A_i \rightarrow \text{Lim } B_i)$, assuming the limits exist.

Proof. Chase the diagrams and obtain those commute diagrams:

$$\begin{array}{ccccc} & & i & & \\ & & \downarrow & \searrow & \\ \text{Ker } h_i & \xrightarrow{\quad} & A_i & \xrightarrow{\quad} & B_i \\ \uparrow & \nearrow \exists! & \uparrow & \uparrow & \uparrow \\ \text{Lim Ker } h_i & \xrightarrow{\quad} & \text{Lim } A_i & \xrightarrow{\exists!} & \text{Lim } B_i \\ \downarrow \exists! & \searrow & \downarrow & \searrow & \downarrow \\ \text{Ker}(\text{Lim } A_i \rightarrow \text{Lim } B_i) & \xrightarrow{\quad} & 0 & \xrightarrow{\quad} & 0 \end{array}$$

$$\begin{array}{ccccccc} & & C & & & & \\ & & \downarrow & \searrow & \searrow & \searrow & \\ & & \text{Ker } h_i & \xrightarrow{\quad} & A_i & \xrightarrow{\quad} & B_i \\ \uparrow & \nearrow \exists! & \downarrow \exists! & \downarrow & \downarrow & \downarrow & \\ \text{Ker}(\text{Lim } A_i \rightarrow \text{Lim } B_i) & \xrightarrow{\quad} & \text{Ker } h_j & \xrightarrow{\quad} & A_j & \xrightarrow{\quad} & B_j \end{array}$$

□

Exercise 5.K

Make sense of the statement that “limits commute with limits” in a general category, and prove it. (Hint: recall that kernels are limits. The previous exercise should be a corollary of this one.)